7.12.5 Treatment and Preventive Measure

In early syphilis

a. Benzathine benzyl penicillin, 24 lakhs units intramuscularly in a single dose.

b. Alternatively, doxycycline 100 mg twice a day orally for 15 days.

In late syphilis

Benzathine benzyl penicillin 24 lakhs units, intramuscularly once weekly for 3 weeks.

- Avoiding sexual contact with an infected individual.
- Use of sex barriers (condoms).

7.13 Leptospira Interrogans

Spirochaetes of the genus *Leptospira* are actively motile, delicate and possess numerous closely wound spirals with characteristic hooked ends. Several Leptospires are saprophytes, while many are potential pathogens of rodents, domestic animals and humans. The genus Leptospira consists of two important species, which are *Leptospira interrogans* and *Leptospira biflexa.*

Leptospira interrrogans is the causative agent of leptospirosis, a zoonotic disease. The word Leptospira is derived from Latin word 'Leptos' = fine or thin and 'spira' = Coil and interrogans = Question mark (The shape of this spirochete accounts for its name)

7.13.1 Morphology

- They are spiral bacteria (5–20µm × 0.1µm) with numerous closely set coils. Their ends are hooked and resemble umbrella handles.
- They are actively motile by rotatory movements.
- They cannot be seen under light microscope due to its thinness, best observed by dark fieldmicroscopy (Figure 7.24), phase contrast and electron microscope.
- They stain poorly with aniline dyes, it may be stained with giemsa stain or silver impregnation techniques.

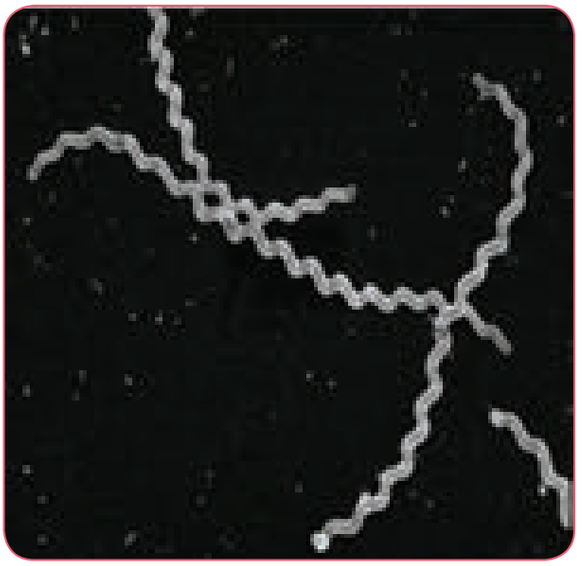


Figure 7.24: Dark field microscopy of *Leptospira interrogans*

7.13.2 Antigenic Structure

Leptospires show considerable antigenic cross reaction.

a. Genus – Specific somatic antigen – It is present in all members of the genus.

b. Surface antigens – This antigen is used to classify Leptospira into serogroups and serotypes.

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